Multimedia Tools and Applications (2019) 78:19905–19916
https://doi.org/10.1007/s11042-019-7327-8

A healthcare monitoring system using random forest and internet of things (IoT)

Pavleen Kaur[1] · Ravinder Kumar[1] · Munish Kumar[2]

Received: 14 May 2018 / Revised: 21 December 2018 / Accepted: 4 February 2019 /
Published online: 22 February 2019

Abstract

The Internet of Things (IoT) enabled various types of applications in the field of information technology, smart and connected health care is notably a crucial one is one of them. Our physical and mental health information can be used to bring about a positive transformation change in the health care landscape using networked sensors. It makes it possible for monitoring to come to the people who don't have ready access to effective health monitoring system. The captured data can then be analyzed using various machine learning algorithms and then shared through wireless connectivity with medical professionals who can make appropriate recommendations. These scenarios already exist, but we intend to enhance it by analyzing the past data for predicting future problems using prescriptive analytics. It will allow us to move from reactive to visionary approach by rapidly spotting trends and making recommendations on behalf of the actual medical service provider. In this paper, the authors have applied different machine learning techniques and considered public datasets of health care stored in the cloud to build a system, which allows for real time and remote health monitoring built on IoT infrastructure and associated with cloud computing. The system will be allowed to drive recommendations based on the historic and empirical data lying on the cloud. The authors have proposed a framework to uncover knowledge in a database, bringing light to disguise patterns which can help in credible decision making. This paper has evaluated prediction systems for diseases such as heart diseases, breast cancer, diabetes, spect_heart, thyroid, dermatology, liver disorders and surgical data using a number of input attributes related to that particular disease. Experimental results are conducted using a few machine learning algorithms considered in this paper like K-NN, Support Vector Machine, Decision Trees, Random Forest, and MLP.

Keywords Internet of things · Data mining · Machine learning · Healthcare

✉ Munish Kumar
munishcse@gmail.com

Extended author information available on the last page of the article

1 Introduction

Machine Learning is considered as a branch of artificial intelligence. Machine learning plays an important role in diagnosis of many diseases like cancer since medical data has become digitized. Researchers have explored many machine leaning techniques in the field of health care system in the past decade. These techniques work as (a) segmentation or detection of abnormalities and (b) classified the segmented abnormality into malignant. Both phases can be cast as a supervised classification task, first providing semantic information by classifying each pixel/voxel as belonging to a suspicious lesion or not, and second permitting to further analyze or quantify the detected/segmented abnormalities and to finally classify them as dangerous or not. Internet of Things (IoT) is a genuine innovation empowered network of networks. As IoT has brought everything associated through the web, this associated organization offers new chances to improve activities crosswise over assembling, farming, the economy and restorative administrations. Out of all, medical services represent one of the social and economic difficulties that each nation faces. Human services executives, clinicians, analysts, and other wellbeing experts are challenging strain for adopting to developing requirements from both the general society and the private division. As the quantity of delicate individuals, for example, seniors, impaired and those with unending sicknesses living in their homes, or remote areas are expanding exponentially. So, the need for remote patient observing is turning into a standard practice. The analysis is presently on how to give appropriate and quality care outside the healing facility condition. In this context, progressed electronic medicinal service administrations are required to be made accessible through a system whenever, wherever and to anybody. For this IoT has been proposed as a framework for restorative sensor correspondence infrastructure which will make utilization of remote innovations that empower the continuous transmission of information about a patient's condition to caregivers. Various versatile gadgets are accessible that can distinguish certain medical conditions—heart rate, blood pressure, breath, liquor level, etc. from a user's touch and exchange those accounts through the cell phones. At that point, IoT with the help of interconnected sensors using mobile devices will accommodate continuous checking of patient health status and deal with their treatment from inaccessible areas. In this manner, it is normal that IoT will assume a noteworthy part in next – age health service provisioning.

However, this development and deployment of the healthcare system through cell phones presents a few difficulties, some of them being marvelous information storing and administration, interoperability, accessibility of heterogeneous assets, security and protection, data anomaly and so on. Cloud computing gives the facility to get to shared assets and basic foundation on an individual and straightforward way, offering services according to request over the system and performing tasks that are as indicated by the progressing needs. IoT based health care monitoring framework includes "internet of wellbeing sensor things". These things create colossal volumes of data that couldn't be taken care of by a doctor. The doctor's fundamental concern is that he has to settle on choices about patient's wellbeing for which he needs to isolate the data around one specific patient from the flood of health data corresponding to the huge number of patients. Along these lines, an IoT operator will be utilized to send health data into the cloud and the cloud would deal with the huge amount of data and give suitability to big data analytics. Continuous alarming of the patient's health is empowered with this information mining and analysis.

The patient's data (such as blood pressure, temperature, and blood glucose etc.) is collected using different body sensor nodes. Further, the collected data is communicated through IoT agent/mobile devices on the cloud network. In the cloud network, data is processed and stored in an efficient manner. Doctors take action remotely by processing the data and uploading their responses onto the cloud. A primary project dealing with healthcare initiative is the quality of service for diagnosing disease capability and performing a service of effective treatments to patients. Diagnosis is complicated but essential undertaking needs to be executed efficaciously. The prediction is made frequently based on the doctor's experience and understanding which may at times be erroneous leading to undesirable effects. Therefore, there arises a need of an automated medical analysis system to be designed that will take gain of collected information base and decision support system. This system can assist in diagnosing disorder much frequently with lesser medical tests and even before major symptoms are felt by the patient. For the realization of this need, authors make use of the healthcare information systems managed by hospitals. They collected huge amounts of data in order to extract hidden information and build an intelligent disease prediction system that diagnose disease using a historical database of various diseases. Therefore, machine learning techniques are adopted for processing large amount of patient data and considerably extract inherent, formerly concealed and substantially useful information about data in an effective way in the e-healthcare network. Tao et al. [10] have exhibited a new classifier named multicolumn BLSTM (MBLSTM), which viably joins diverse different acceleration signal features to additionally enhance action recognition precision. They likewise inferred that cell phones based human action dataset propose that MBLSTM is better than other cutting edge strategies. Tao et al. [11] have planned a structure for powerfully taking in the semantic content of mobile images, in light of which the important labels are alloted to the query images. Their proposed structure is extremely obtuse to the noise and the anomalies, and is improved by a half-quadratic enhancement system. They have likewise presumed that their image labeling approach is more robust than the other sparse coding methods. The aim of this paper was to evaluate the performance of existing machine learning techniques for various public dataset for health care applications and to present an efficient methodology for predicting the disease based on IoT network. The paper has been outlined as follows. Section 2 defines the related work done in the health care for diseases of prediction. Section 3 presents, proposed prediction model for detecting a number of diseases using machine learning techniques and IoT. Significance of the proposed work is presented in Section 4. Section 5 depicts the experimental results and performance analysis. Finally, concluding notes of present work are given in Section 6.

2 Related work

Knowledge of medical field is based on the experience or evidence that accumulated through medical experts. The human body is like a complicated machinery which contains a number of components and that can be affected by many factors. Therefore, the modeling of its dis-functions or function is a tedious task. Medical data can be used to make observations and predictions to the diagnosis of various diseases. Machine learning techniques provide better facilitates to design such tools which can learn from previous experience and make predictions that very useful in e-health applications. A few scholars have been presented work for E-health care application. For example, Hameed et al. [5] have presented e-healthcare service framework based on cloud in Iraq, in which all which all the patient data is chronicled in a focal database.. The proposed framework depends on Service Oriented Architecture (SOA) and offers different offices like enhancing cost the executive's time, stores patients profile and takes the correct specialist choice. Parekh and Saleena [9] have presented a cloud based framework for the health care system by using clustering as data mining technique. This methodology empowered patients to get required wellbeing administrations through a versatile application with a few ticks. Vijayarani and Dhayanand [16] used Naïve Bayes and Support Vector Machine (SVM) algorithm to predict kidney disease. The performance analysis is done based on the accuracy and execution time. The experimental results have shown that the SVM technique provides better accuracy compared to Naïve Bayes. Hsu et al. [6] proposed a model based on health information for breast cancer risk assessment. In this method, sampling and dimension reduction techniques were applied to pre-process the testing data. Thereafter, various classifiers were used for risk prediction. Devi and Shyla [2] applied data mining techniques for early prediction of diabetes disease. They used 768 instances collected from PIMA Indian dataset to determine accuracy. Their analysis proves that J48 classifier has better accuracy as compared to other techniques. Turanoglu-Bekar et al. [13] used various decision tree data mining techniques (such as NB tree, J48, LADTree, BFTree, LMT, Random Forest etc.) for prediction of the thyroid disease. The performance analysis is done based on six performance metrics such as accuracy, MAE, PRE, REC, FME, and Kappa statistics. Their results reflected that NB-Tree has higher accuracy as compared to other algorithms.

Verma et al. [15] have proposed Coronary Artery Disease (CAD) method using particle swarm intelligence and K-means algorithm for risk factor identification. They deployed various learning algorithms such as Multi-Layer Perceptron (MLP), Multinomial Logistic Regression (MLR), fuzzy unordered rule induction algorithm and C4.5 for extracting data events. The dataset is collected from the Department of Cardiology, Indira Gandhi Medical College, Shimla, India. This data set consists of 26 features and 335 instances. The experimental results show that MLR achieved a highest accuracy of 88.4%. Forkan et al. [4] have proposed a predictive model ViSiBiD for daily monitoring and disease prevention based on analyzing the vital signs of patients. The machine learning techniques along with some Map-Reduce implementation have been used for learning the cloud platform. They used 4893 patient's dataset which is publicly available and observed that six bio-signals deviated from the normal and different features for observing the data events. Data events are collected in 1–2 h' intervals. Their results show that random forest has the best accuracy of 95.85% as compared to other techniques. Jahangir et al. [7] designed an application of automatic Multi-Layer Perceptron (AutoMLP) for diabetes prediction. In this technique, enhanced class outlier detection is also used. It performs parameter tuning automatically during training process. The outlier detection is performed during pre-processing of data. Osman and Aljahdali [8] proposed a technique to diagnosis diabetes disease based on improved feature extraction. For classification, they considered SVM and K-means techniques. The experimental analysis depicts that the proposed technique provides better classification rate of diabetes diagnosis than existing techniques. Zhang et al. [17] designed a novel statistical learning theory based cancer prediction technique. The technique is capable to apply on binary classification and multi-class problems. The SVM technique creates the large hyperplanes in high dimensional space, which maximizes the separation between data points, and support vectors are used to create hyper-plane. The SVM provides better accuracy but expensive in terms of computational time. From the literature review, it is observed that the use of meta-heuristic techniques is ignored by most of the existing researchers to improve the classification rate of existing machine learning techniques. So, in this paper, efficient machine learning algorithm Random Forest has been used to improve the classification results.

3 Proposed system

The Internet of Things (IoT) is a true technology enabled a network of networks. IoT's is able to connect remote and mobile things or machines through the use of wireless communications for sensors which are low in cost for assisted by computing and storage devices. As IoT has brought everything associated through web, this associated system offers new chances to improve activities across manufacturing, agriculture, economy and medicinal services. IoT based health care monitoring system comprises of "internet on health sensor things". It contains big data that is difficult to maintain by the physician. However, physicians require to utilize this historical data to predict the health of the patients. Many machine learning techniques have been used in the past few decades for health care applications. However, standard machine learning techniques suffer from parameter tuning issue. Therefore, efficient tuning of these parameters has an ability to improve the performance of existing machine learning techniques to predict various medical applications such as cancer, diabetes, brain tumor, etc. The way towards investigating through the information to find hidden associations and predict future patterns has a long history. Once in a while referred to as, "learning revelation in databases". Be that as it may, its establishment involves three inter-mixed disciplines: statistics, artificial intelligence and machine learning. Using these three disciplines, the authors plan to produce a framework which allows for electronic remote health monitoring built on IoT basis and helped with cloud computing. Such situations as of now exist, one can expect to include the advantages of machine learning to the cloud, holding the medicinal services database with a specific end goal to give constant observing of patient status and oversee restorative treatment to each client of the framework. Authors proposed a framework of decision support system in the cloud and ground of machine learning techniques that can be applied on cloud data and check their performances. Patients range from those equipped with all easily available and affordable sensors at their respective homes to the ones situated at remote locations or at a distance from the medical provisioning. The third kind of service seekers includes all the patients visiting the clinics or laboratories which do not have the physical presence of the doctors, but are equipped with all the necessary medical devices and assisting staff who act as the intermediate. They deploy the healthcare information collected from patients using the sensor systems onto the cloud for the doctors to access and respond. All the collected health care information is transferred initially into the mobile devices through a sensor network. The sensor network may be bluetooth, Wi-Fi application or USB based connection. Cell phones go about as the IoT operator and are utilized to send the health information of a patient onto the cloud. The cloud would deal with the expanding volume of health information, creatively share the data crosswise over social insurance frameworks and give reasonability of information mining, which is an important and crucial exercise in performing work. The implementation so far in this research involves application of major machine learning techniques, namely, K-NN, Linear-SVM, Decision Tress, Random Forest and MLP on the dataset of the diseases, namely, breast cancer, diabetes, heart diseases, spect_heart, thyroid, Surgery data, dermatology and liver disorder. Random forest is unexcelled in accuracy among other existing supervised learning algorithms for classification and runs efficiently on large databases. Random forest classifier creates a set of decision trees from a randomly selected subset of the training set. It then aggregates the votes from different decision trees to decide the final class of the test object. In present work, Random forest classifier is used to extract the exact information from the database related to the query dataset of the diseases. The block diagram of the proposed system is illustrated in Fig. 1. The results of the techniques are compared based on their accuracy and Area Under Curve (AUC) as shown in Figs. 2-3.

4 Significance of the proposed system

A few data mining techniques, namely, k-Nearest Neighbor, Linear-SVM, Decision Tree, Random Forest and MLP are used and the performance of these techniques is compared to ensure quality of service to the healthcare industry. K-Nearest Neighbor technique is based on nearest neighbor data points finds the unidentified data points and classifies the data points according to the voting system. The K-NN technique is easy to implement, but requires large storage space, sensitive to noise and slow testing procedure [12]. Linear Support Vector Machine technique is based on statistical learning theory. The technique is capable to be applied to binary classification and multi-class problems. The SVM technique creates the large hyper-planes in high dimensional space, which maximizes the separation between data points, and support vectors are used to create a hyperplane. The SVM provides better accuracy, but expensive in terms of computational time

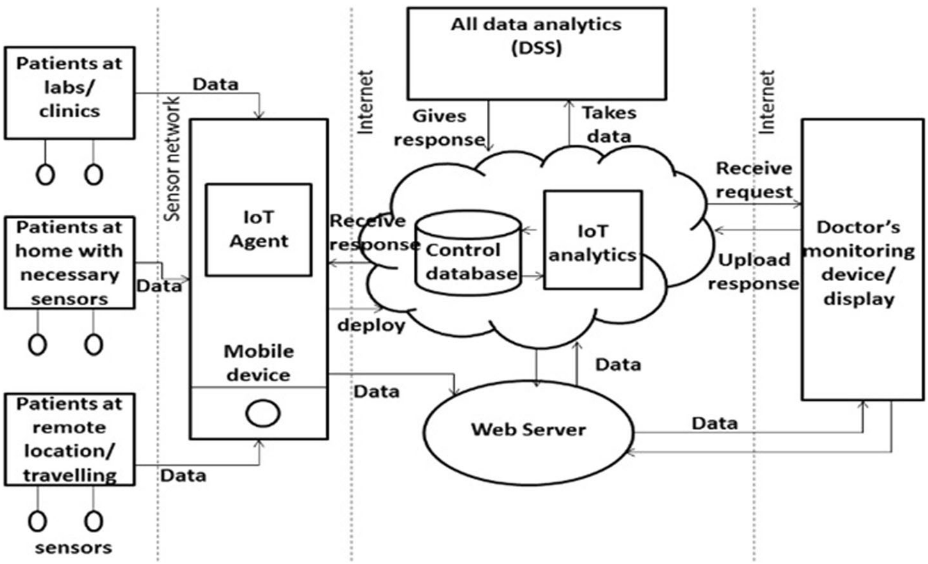

Fig. 1 Block diagram of prediction model for IoT based healthcare network

Table 1 Dataset used for experimental work

Dataset	Class	Number of samples
Breast cancer	2	699
Diabetes	2	768
Heart disease	5	303
Spect_Heart	2	187
Thyroid	6	9172
Surgery_Data	2	470
Dermatology	6	366
Liver_Disorder	2	345

[12]. Decision Tree technique is based on tree-like graph, in which 3 nodes (such as non-leaf, leaf and branch nodes) are used as different attributes for calculating the conditional probabilities. The node at the top is used as the root node, leaf node work as a class label, branch nodes are outcomes of the test, and non-leaf nodes are used to denote the test. The decision tree technique does not require domain knowledge. Also, it is easy to interpret and can handle categorical and numerical data. On the other hand, the performance is dependent on the dataset and restricted to one attribute output [12]. Random Forest is developed by Leo Breiman. The technique is adopted in a number of applications such as classification, prediction, variable study and selection [14]. The technique has several characteristics such as it can be applied to two class or multi-class problem, mixture of continuous and categorical predictors. On the other side, fine-tuning of input parameters is required for better performance [3]. MLP is the feed forward artificial neural network technique. In this technique, neural network layer tries to calculate errors in the output and revert the output to hidden layer to update the internal weight [1]. The MLP technique has a high prediction accuracy as compared to other techniques. The IoT has many health care applications like remote health monitoring, fitness programs, medication at home by healthcare providers, etc. Health care services based on IoT are expected to reduce health diagnostic costs and increase the quality of health care management. Due to distant physical locations, sometimes manual health care services are unable to reach on time and thus causes severe danger to life. So, the main motivation behind using the health care system based on IoT is to use updated health care networks driven by wireless technologies in order to support chronic diseases, early diagnosis, real-time monitoring and medical emergencies. Moreover, the IoT provides efficient scheduling of limited resources by ensuring their best use and services to patients. According to the report given by the United Nations Population Fund, the

Table 2 Comparative study of accuracy achieved with various techniques

Data set	k-NN	Linear-SVM	Decision tree	MLP	Random forest
Breast cancer	95.71%	96.42%	92.85%	95.71%	96.42%
Diabetes	74.67%	79.87%	75.97%	78.57%	81.16%
Heart disease	55.73%	57.37%	52.45%	47.54%	55.73%
Spect_Heart	66.31%	67.91%	66.84%	65.24%	72.19%
Thyroid	61.45%	66.63%	66.57%	66.95%	70.12%
Surgery_Data	78.72%	81.91%	81.90%	73.40%	80.85%
Dermatology	94.52%	95.89%	93.15%	95.90%	97.26%
Liver_Disorder	63.76%	69.56%	59.42%	68.11%	66.67%

Table 3 AUC of different techniques considered in this study

Data set	k-NN	Linear-SVM	Decision tree	MLP	Random forest
Breast cancer	95.3%	96.3%	95.5%	99.4%	99.6%
Diabetes	70.5%	74.9%	77.7%	80.6%	85.2%
Heart disease	72.4%	72.4%	71.3%	78.2%	82.3%
Spect_Heart	66.1%	66.2%	59.6%	67.4%	75.8%
Thyroid	65.8%	67.5%	74.0%	80.7%	81.4%
Surgery_Data	54.9%	49.4%	48.2%	57.9%	57.3%
Dermatology	96.6%	97.5%	95.4%	99.9%	99.9%
Liver_Disorder	64.9%	66.4%	58.2%	80.1%	71.2%

people aged 60 and older make up 12.3% of the global population and by 2050, that number will rise to almost 22%. Thus, in such scenario, there is going to be a huge need of medical assistance to the elderly population, in which IoT based health care system will play a vital role in bridging the gap between the doctor and the patients.

5 Experimental results and discussion

In this section, results of the healthcare system using a few machine learning algorithms like K-NN, Support Vector Machine, Decision Trees, Random Forest, and MLP are presented. Authors have used a number of public datasets for different diseases, namely, breast cancer, diabetes, heart disease, Spect-heart, thyroid, surgery, dermatology and liver disorder as shown in Table 1. These datasets are downloaded from https://archive.ics.uci.edu/ml/datasets.html. For experimental work, the data set of each disease is divided into training and testing dataset i.e. 80% data as the training set and remaining 20% data as the testing set. To assess the performance of various machine learning algorithms, authors have used the Waikato Environment for Knowledge Analysis (WEKA) open source tool. Disease wise comparative analysis of accuracy and Area Under Curve (AUC) for different machine learning techniques are presented in Tables 2 and 3, respectively. These results are graphically also shown in Figs. 2 and 3. In breast cancer related dataset which contains a total of 699 samples corresponding to 2 classes, the highest accuracy of 96.42% has been achieved by two machine learning techniques,

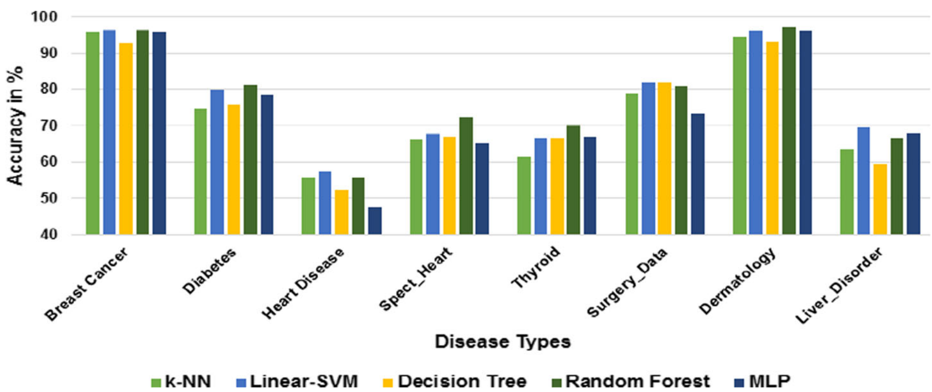

Fig. 2 Accuracy achieved for different diseases

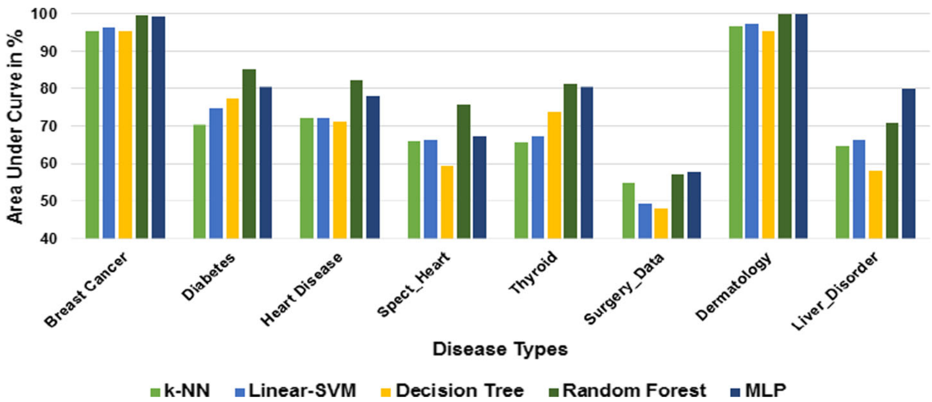

Fig. 3 AUC of different machine learning techniques

namely, Linear SVM and Random forest. Random forest provides highest diagnostic rate of 99.6% using AUC parameter. The dataset related to diabetes contains 768 samples and 2 classes. On diabetes, Random forest provides highest accuracy and diagnostic rate of 81.16% and 85.2%, respectively. Heart disease dataset comprises a total of 303 samples corresponding to 5 classes. The highest accuracy of 57.37% has been achieved on this dataset using Linear-SVM, while Random forest provides the highest diagnostic rate of 82.3%. Spect_Heart dataset contains 187 samples and 2 different classes. In this dataset, Random forest reports the highest accuracy and diagnostic rate of 72.19% and 75.8%, respectively. Data set on thyroid contains a total of 9172 samples and 6 different classes. Random forest provides the highest accuracy and diagnostic rate of 70.12% and 81.4%, respectively, on this dataset. Surgery_Data comprises of 470 samples corresponding to 2 different classes. On this data set, the highest accuracy and diagnostic rate of 81.91% and 57.9% have been achieved by Linear-SVM and MLP, respectively.

Dermatology dataset contains 366 samples and 6 classes. The highest accuracy of 97.26% of this dataset has been reported by Random forest. The highest diagnostic rate of 99.9% has been observed by Random forest and MLP. Liver_Disorder dataset comprises a total of 345 samples corresponding to 2 different classes. In this data set, Linear-SVM provides highest accuracy of 69.56% and MLP provides highest diagnostic rate of 80.1%.

6 Conclusion

The work presented in this paper proposes a health care system based on random forest classifier and IoT. The proposed system will improve interactivity between patients and doctors. Experimental results are conducted using various datasets related to different diseases, namely, breast cancer, diabetes, heart disease, spect-heart, thyroid, surgery, dermatology and liver disorder for testing the effectiveness of the proposed work. The machine learning techniques that have been employed in this work are K-NN, Support Vector Machine, Decision Trees, Random Forest, and MLP. Maximum accuracy of 97.26% has been achieved on Dermatology dataset using Random Forest machine learning technique. It has been analyzed that on an average, random forest provides good accurate results for each of the datasets considered. This accuracy can probably be increased by considering a larger data set while training the proposed system with different machine learning algorithms. Therefore, in the near future, this work can be extended to other applications such as earth observations, weather forecasting, etc. Security of data is the main implication in IoT based framework because IoT devices collect useful data with the help of various existing technologies and transfer that data to other IoT devices or systems.

References

1. Costa K, Ribeiro P, Carmargo A, Rossi V, Martins H, Neves M, Fabris R, Imaisumi R, Papa JP (2013) Comparison of the techniques decision tee and MLP for data mining in SPAMs detection in computer networks. Proceedings of the 3rd international conference on innovative computing technology, pp 344–348
2. Devi MR, Shyla JM (2016) Analysis of various data mining techniques to predict diabetes mellitus. Int J Appl Eng Res 11(1):727–730
3. Diaz-Uriarte R, Alverez-de-Andres S (2006) Gene selection and classification of microarray data using random forest. BMC Bioinformatics. https://doi.org/10.1186/1471-2105-7-3
4. Forkan ARM, Khalil I, Atiquzzaman M (2017) ViSiBiD: a learning model for early discovery and real time prediction of severe clinical events using vital signs as big data. Comput Netw 113:244–257
5. Hameed RT, Mohamad OA, Hamid OT, Tapus N (2015) Design of e-healthcare management system Basedon cloud and service oriented architecture. Proceedings of the 5th IEEE international conference on E-health and bioengineering (EHB), pp 1–4
6. Hsu JL, Hung PC, Lin HY, Hsieh CH (2015) Applying under-sampling techniques and cost-sensitive learning methods on risk assessment of breast cancer. J Med Syst 39(4). https://doi.org/10.1007/s10916-015-0210-x
7. Jahangir M, Afzal H, Ahmed M, Khurshid K, Nawaz R (2017) An expert system for diabetes prediction using auto tuned multi-layer perceptron. Proceedings of the intelligent system conference, pp 722–728
8. Osman AH, Aljahdali HM (2017) Diabetes disease diagnosis method based on feature extraction using k-svm. Int J Adv Comput Sci Appl 8(1):236–244
9. Parekh M, Saleena B (2015) Designing a cloud based framework for healthcare system and applying clustering techniques for region wise diagnosis. 2nd international symposium on big data and cloud computing (ISBCC'15), 50:537–542
10. Tao D, Wen Y, Hong R (2016) Multicolumn bidirectional long short-term memory for mobile devices-based human activity recognition. IEEE Internet Things J 3(6):1124–1134
11. Tao D, Cheng J, Gao X, Li X, Deng C (2017) Robust sparse coding for Mobile image labeling on the cloud. IEEE Trans Circuits Syst Video Technol 27(1):62–72
12. Tomar D, Agarwal S (2013) A survey on data mining approaches for healthcare. International Journal of Bio-Science and Bio-Technology 5(5):241–266
13. Turanoglu-Bekar E, Ulutagay G, Kantarc-Savas S (2016) Classification of thyroid disease by using data mining models: a comparison of decision tree algorithm. Oxford Journal of Intelligent Decision and Data Science 2:13–28
14. Verikas A, Gelzinis A, Bacauskiene M (2011) Mining data with random forest: a survey and results of new tests. Pattern Recogn 44(2):330–349
15. Verma L, Srivastava S, Negi PC (2016) A hybrid data mining model to predict coronary artery disease cases using non-invasive clinical data. J Med Syst 40(7):178
16. Vijayarani S, Dhayanand S (2015) Data mining classification algorithms for kidney diseases prediction. International Journal on Cybernetics & Informatics 4(4):13–25
17. Zhang L, Zhou W, Wang B, Zhang Z, Li F (2018) Applying 1-norm svm with squared loss to gene selection for cancer classification. Appl Intell 48(7):1878–1890

Pavleen Kaur graduated in Computer Science & Engineering from Thapar Institute of Engineering and Technology, Patiala, India in 2014. Presently, she is pursuing her Masters in Software Engineering from Thapar Institute of Engineering and Technology, Patiala, India. Her industrial experience includes functioning as a Business Analyst intern and her current examination works incorporate use of Deep Learning in Machine Translation and Big Data Mining.

Ravinder Kumar received the Ph.D. degree in computer science and engineering from Thapar University in 2015. He is currently an Assistant Professor with the Computer Science and Engineering Department, Thapar University. He has already developed a complete working project on speech recognition and handwritten recognition for Indian regional language (Punjabi). His area of research includes theoretical and practical aspects of combinatorial optimization, approximation algorithm, and mathematical programming. He is the member of various professional bodies and serves as a Reviewer to many referred journals.

Munish Kumar received his Master's degree in Computer Science & Engineering from Thapar University, Patiala, India in 2008. He received his Ph.D. degree in Computer Applications from Thapar University, Patiala, India in 2015. He started his career as an Assistant Professor in computer application at Jaito Centre of Punjabi university, Patiala. Presently, he is working as Assistant Professor in Department of Computational Sciences, Maharaja Ranjit Singh Punjab Technical University, Bathinda, Punjab, India. His research interests include Character Recognition, Computer Vision and Pattern Recognition.

Affiliations

Pavleen Kaur[1] **· Ravinder Kumar**[1] **· Munish Kumar**[2]

Pavleen Kaur
amicablepavleen@gmail.com

Ravinder Kumar
ravinder@thapar.edu

1 Computer Science and Engineering Department, Thapar Institute of Engineering & Technology, Patiala, Punjab, India

2 Department of Computational Sciences, Maharaja Ranjit Singh Punjab Technical University, Bathinda, Punjab, India